ASSIGNMENT

1 If $a+\frac{1}{a}=5$,the find

$(i)a^2+\frac{1}{a^2}$ $(ii)a^3+\frac{1}{a^3}$ $(iii)a^5+\frac{1}{a^5}$

Solution:

$$a^2+\frac{1}{a^2}=23 \ \& a^3+\frac{1}{a^3}=110$$

$$a^5+\frac{1}{a^5}=23\times10-5=2530-5=2525$$

2 If $a+\frac{1}{a}=m$ & $a-\frac{1}{a}=n$ then find the relation b/w m & n

Solution:

$$\left(a+\frac{1}{a}\right)^2-\left(a-\frac{1}{a}\right)^2=m^2-n^2$$

$$4=m^2-n^2$$

3 If $a^2+b^2+c^2=10$ and a + b + c = 4 , find ab + bc + ca

Solution:

$$(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ac)$$

$$16=10+2x$$

X = 3

4 If 4x + 3y = 4 &xy = 2 then find $64x^3+27y^3$

Solution:

$$64x^3+27y^3=(4x+3y)(16x^2+9y^2-12xy)$$

$$=4\times\left((4x+3y)^2-36xy\right)$$

$$=4\times(16-72)=-224$$

5 If a + b + c = 6 and ab + bc + ac = 12 find the value of $a^3+b^3+c^3-3abc$

Solution:

$$\left(a+b+c\right)^2=a^2+b^2+c^2+2\left(ab+bc+ca\right)$$

$$36=a^2+b^2+c^2+24$$

$$a^2+b^2+c^2=12$$

$$a^3+b^3+c^3-3abc=\left(a+b+c\right)\left(12-12\right)=0$$

6 Simplify : $\dfrac{\left(a^2-b^2\right)^3+\left(b^2-c^2\right)^3+\left(c^2-a^2\right)^3}{\left(a-b\right)^3+\left(b-c\right)^3+\left(c-a\right)^3}$

Solution:

$$\frac{3\left(a^2-b^2\right)\left(b^2-c^2\right)\left(c^2-a^2\right)}{3\left(a-b\right)\left(b-c\right)\left(c-a\right)}=\left(a+b\right)\left(b+c\right)\left(c+a\right)$$

7 If $\dfrac{1}{a-b}+\dfrac{1}{b-c}+\dfrac{1}{c-a}=3$, find the value of $\dfrac{1}{\left(a-b\right)^2}+\dfrac{1}{\left(b-c\right)^2}+\dfrac{1}{\left(c-a\right)^2}$

Solution:

$$\left(\frac{1}{a-b}+\frac{1}{b-c}+\frac{1}{c-a}\right)^2=\frac{1}{\left(a-b\right)^2}+\frac{1}{\left(b-c\right)^2}+\frac{1}{\left(c-a\right)^2}+0=9$$

8 Given $x+y=\dfrac{5}{2}, x^2+y^2=\dfrac{13}{4}$, find the value of x^5+y^5.

Solution:

$$\left(x+y\right)^2-x^2-y^2=2xy$$

$$\frac{25}{4}-\frac{13}{4}=2xy\Rightarrow xy=\frac{3}{2}$$

$$x^3+y^3=\left(\frac{5}{2}\right)^3-3\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)$$

$$\frac{125}{8}-\frac{45}{4}=\frac{35}{8}$$

$$x^5+y^5=\left(x^3+y^3\right)\left(x^2+y^2\right)-x^2y^2\left(x+y\right)$$

$$=\left(\frac{35}{8}\right)\left(\frac{13}{4}\right)-\left(\frac{3}{2}\right)^2\left(\frac{5}{2}\right)$$

$$\frac{455-45\times 4}{32}=\frac{455-180}{32}=\frac{275}{32}$$

9 If a,b,c,d> 0 and $a^4+b^4+c^4+d^4=4abcd$, prove that a = b = c = d.

Solution:

We rewrite the given equality in the form $a^4+b^4+c^4+d^4-4abcd=0$, and use the technique for completing squares, then

$$0 = a^4+b^4+c^4+d^4-4abcd$$

$$= \left(a^4-2a^2b^2+b^4\right)+\left(c^4-2c^2d^2+d^4\right)+\left(2a^2b^2+2c^2d^2-4abcd\right)$$

$$=\left(a^2-b^2\right)^2+\left(c^2-d^2\right)^2+2\left(ab-cd\right)^2$$

Therefore $a^2-b^2=0, c^2-d^2=0, ab-cd=0.$ Since a,b,c,d> 0, so a = b, c = d, and $a^2=c^2$,i.e. a = c. Thus a = b = c = d.

10 If $x=2+2^{\frac{2}{3}}+2^{\frac{1}{3}}$ then $x^3-6x^2+6x=?$

Option

(a) 3 (b) 2 (c) 1 (d) None

Answer: (b)

Solution:

Cubing both sides

$$\left(x-2\right)^3=\left(2^{\frac{2}{3}}+2^{\frac{1}{3}}\right)^3$$

$$x^3-8-6x\left(x-2\right)=\left(2^{\frac{2}{3}}\right)^3+\left(2^{\frac{1}{3}}\right)^3+3\left(2\right)^{\frac{2}{3}}\left(2^{\frac{1}{3}}\right)\left(2^{\frac{2}{3}}+2^{\frac{1}{3}}\right)$$

$$x^3-8-6x^2+12x=4+2+3(2)(x-2)$$

$$x^3-6x^2+12x-8=6+6x-12$$

$$x^3-6x^2+6x=8+6-12$$

$$x^3-6x^2+6x=2$$

11 Factorize $\left(d^2-c^2+a^2-b^2\right)^2-4\left(bc-da\right)^2$

Solution:

$$\left(d^2-c^2+a^2-b^2\right)^2-4\left(bc-da\right)^2=\left(d^2-c^2+a^2-b^2\right)^2-\left(2bc-2da\right)^2$$

$$=\left(d^2-c^2+a^2-b^2-2bc+2da\right)\left(d^2-c^2+a^2+b^2+2bc-2da\right)$$

$$=\left[\left(d+a\right)^2-\left(b+c\right)^2\right]\left[\left(d-a\right)^2-\left(b-c\right)^2\right]$$

$$=\left(d+a-b-c\right)\left(d+a+b+c\right)\left(d+b-a-c\right)\left(d+c-a-b\right)$$

12 Factorize $\left(d^2 - c^2 + a^2 - b^2\right)^2 - 4\left(bc - da\right)^2$

Solution:

$$64x^6 - 729y^{12} = (2x)^6 - \left(3y^2\right)^6 = \left[(2x)^3 - \left(3y^2\right)^3\right]\left[(2x)^3 + \left(3y^2\right)^3\right]$$

$$= \left(2x - 3y^2\right)\left[\left(2x^2\right) + (2x)\left(3y^2\right) + \left(3y^2\right)^2\right]\left(2x + 3y^2\right)\left[(2x)^2 - (2x)\left(3y^2\right) + \left(3y^2\right)^2\right]$$

$$= \left(2x + 3y^2\right)\left(2x + 3y^2\right)\left(4x^2 + 6xy^2 + 9y^4\right)\left(4x^2 - 6xy^2 + 9y^4\right)$$

13 Factorize (i) $x^4 + 2x^3 + 7x^2 + 6x - 7$ (ii) $x^3 + 9x^2 + 23x + 15$

Solution:

Let $y = x^2 + x$. Then

(i) $x^4 + 2x^3 + 7x^2 + 6x - 7 = x^2\left(x^2 + x\right) + x\left(x^2 + x\right) + 6\left(x^2 + x\right) - 7$

$$= \left(x^2 + x + 6\right)\left(x^2 + x\right) - 7 = y^2 + 6y - 7 = (y + 4)(y - 1)$$

$$= \left(x^2 + x + 7\right)\left(x^2 + x - 1\right)$$

(ii) $x^3 + 9x^2 + 23x + 15 = x^2(x + 1) + 8x(x + 1) + 15(x + 1)$

$$= (x + 1)\left(x^2 + 8x + 15\right) = (x + 1)(x + 3)(x + 5)$$

14 Factorize (a + 1) (a + 2) (a + 3) (a + 4) – 120

Solution:

$$(a + 1)(a + 2)(a + 3)(a + 4) - 120$$

$$= \left[(a + 1)(a + 4)\right]\left[(a + 2)(a - 3)\right] - 120$$

$$= \left(a^2 + 5a + 4\right)\left(a^2 + 5a + 6\right) - 120$$

$$= \left[\left(a^2 + 5a + 5\right) - 1\right]\left[\left(a^2 + 5a + 5\right) + 1\right] - 120$$

$$= \left(a^2 + 5a + 5\right)^2 - 121 = \left(a^2 + 5a + 5\right)^2 - 11^2$$

$$= \left(a^2 + 5a - 6\right)\left(a^2 + 5a + 16\right) = (a - 1)(a + 6)\left(a^2 + 5a + 16\right)$$

15 If $x^2 + 4y^2 - 3x + 9 = 6y + 2xy$ then the value of x + 2y

Solution:

$$(x - 2y)^2 + (2y - 3)^2 + (x - 3)^2 = 0$$

$$x = 2y = 3$$

$x+2y=6$

16 Factoriz: $x^{11}+x^{10}+\ldots+x^2+x+1$

Solution:

$$x^{11}+x^{10}+\ldots+x^2+x+1$$

$$\frac{x^{12}-1}{x-1}=\frac{\left(x^6-1\right)\left(x^6+1\right)}{\left(x-1\right)}=\frac{\left(x^3-1\right)\left(x^3+1\right)\left(x^6+1\right)}{x-1}$$

$$=\frac{\left(x-1\right)\left(x^2+x+1\right)\left(x+1\right)\left(x^2-x+1\right)\left(x^6+1\right)}{x-1}$$

$$=\left(x+1\right)\left(x^2-x+1\right)\left(x^2+x+1\right)\left(x^6+1\right)$$

$$=\left(x+1\right)\left(x^2-x+1\right)\left(x^2+x+1\right)\left(x^2+1\right)(x^4-x^2+1)$$

17 If $M=3x^2-8xy+9y^2-4x+6y+13$ (where x, y are real numbers), then M must be

Option

(a) Positive (b) Negative (c) 0; (d) an integer

Answer: (c)

Solution:

$$M=2\left(x^2-4xy+4y^2\right)+x^2-4x+4+y^2+6y+9$$

$$=2\left(x-2y\right)^2+\left(x-2\right)^2+\left(y+3\right)^2>0$$

18 Given $a+b=c+d$ and $a^2+b^2=c^2+d^2$. Prove that $a^{2009}+b^{2009}=c^{2009}+d^{2009}$.

Solution:

$$a+b=c+d \quad \ldots \quad (1)$$

$$a^2+b^2+2ab=c^2+d^2+2cd$$

$$ab=cd \quad (2)$$

$$a^2+b^2-2ab=c^2+d^2-2cd$$

$$a-b=\pm\left(c-d\right) \quad (3)$$

From 1 and 3

$2a=2c$

$a=c$

$b=d$

or $2a=2d$

$a=d$

$b=c$